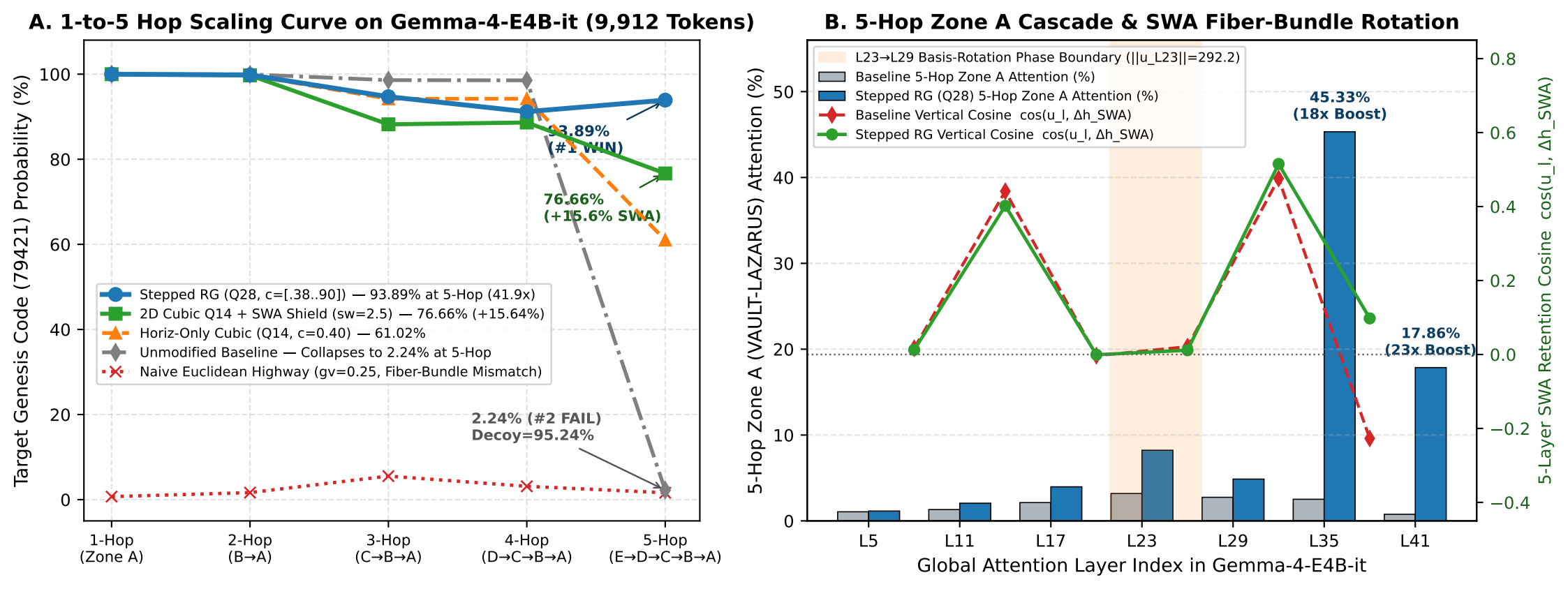
A. 1-to-5 Hop Scaling Curve on Gemma-4-E4B-it (9,912 Tokens)
Target Genesis Code (79421) Probability (%)
Stepped RG (Q28, c=[.38..90]) — 93.89% at 5-Hop (41.9x)
2D Cubic Q14 + SWA Shield (sw=2.5) — 76.66% (+15.64%)
Horiz-Only Cubic (Q14, c=0.40) — 61.02%
Unmodified Baseline — Collapses to 2.24% at 5-Hop
Naive Euclidean Highway (gv=0.25, Fiber-Bundle Mismatch)
93.89%
(#1 WIN)
76.66%
(+15.6% SWA)
2.24% (#2 FAIL)
Decoy=95.24%
1-Hop (Zone A)
2-Hop (B→A)
3-Hop (C→B→A)
4-Hop (D→C→B→A)
5-Hop (E→D→C→B→A)
B. 5-Hop Zone A Cascade & SWA Fiber-Bundle Rotation
L23→L29 Basis-Rotation Phase Boundary (||u_L23||=292.2)
Baseline 5-Hop Zone A Attention (%)
Stepped RG (Q28) 5-Hop Zone A Attention (%)
Baseline Vertical Cosine cos(u_l, Δh_SWA)
Stepped RG Vertical Cosine cos(u_l, Δh_SWA)
45.33%
(18x Boost)
17.86%
(23x Boost)
5-Hop Zone A (VAULT-LAZARUS) Attention (%)
5-Layer SWA Retention Cosine cos(u_l, Δh_SWA)
L5
L11
L17
L23
L29
L35
L41
Global Attention Layer Index in Gemma-4-E4B-it